

8.4.32

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Question: The line passing through the extremity **A** of the major axis and extremity **B** of the minor axis of the ellipse $x^2 + 9y^2 = 9$ meets its auxiliary circle at the point **M**. Then the area of the triangle with the vertices at **A**, **M**, and the origin **O** is (2009)

1) $\frac{31}{10}$

2) $\frac{29}{10}$

3) $\frac{21}{10}$

4) $\frac{27}{10}$

Solution

The solution is derived using a linear algebra approach by representing the geometric elements as vectors and matrices.

The given equation of the ellipse is $x^2 + 9y^2 = 9$. We can express this in the standard matrix form for a conic, $\mathbf{x}^T \mathbf{V} \mathbf{x} + f = 0$. Let the position vector $\mathbf{x}$ be defined as:

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (1)$$

The equation can then be written as:

$$\mathbf{x}^T \begin{pmatrix} 1 & 0 \\ 0 & 9 \end{pmatrix} \mathbf{x} - 9 = 0 \quad (2)$$

Here, the matrix $\mathbf{V} = \begin{pmatrix} 1 & 0 \\ 0 & 9 \end{pmatrix}$ and the scalar $f = -9$.

To find the semi-axes, we normalize the equation to the form $\mathbf{x}^T \mathbf{A} \mathbf{x} = 1$:

$$\frac{x^2}{9} + \frac{y^2}{1} = 1 \implies \mathbf{x}^T \begin{pmatrix} 1/9 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} = 1 \quad (3)$$

The semi-axis lengths are the reciprocal of the square roots of the eigenvalues of the matrix $\mathbf{A} = \begin{pmatrix} 1/9 & 0 \\ 0 & 1 \end{pmatrix}$.

- The eigenvalues are $\lambda_1 = 1/9$ and $\lambda_2 = 1$.
- The major semi-axis is $a = 1/\sqrt{\lambda_1} = 3$. The corresponding eigenvector is $\mathbf{v}_a = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
- The minor semi-axis is $b = 1/\sqrt{\lambda_2} = 1$. The corresponding eigenvector is $\mathbf{v}_b = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

The vertices **A** and **B** are at the extremities of these axes.

- Position vector of **A** : $\mathbf{a} = a \cdot \mathbf{v}_a = 3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$. So, **A** = (3, 0).
- Position vector of **B**: $\mathbf{b} = b \cdot \mathbf{v}_b = 1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. So, **B** = (0, 1).

The vector equation of the line passing through **A** and **B** is:

$$\mathbf{x}(t) = \mathbf{a} + t(\mathbf{b} - \mathbf{a}) = \begin{pmatrix} 3 \\ 0 \end{pmatrix} + t \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right) = \begin{pmatrix} 3 - 3t \\ t \end{pmatrix} \quad (4)$$

The auxiliary circle has the equation $x^2 + y^2 = a^2$, which is $x^2 + y^2 = 9$. We find the intersection point **M** by substituting the parametric equations of the line into the circle's equation:

$$(3 - 3t)^2 + t^2 = 9 \quad (5)$$

$$9 - 18t + 9t^2 + t^2 = 9 \quad (6)$$

$$10t^2 - 18t = 0 \implies 2t(5t - 9) = 0 \quad (7)$$

The solutions are $t = 0$ (which corresponds to point **A**) and $t = 9/5$. The position vector of **M** is found using $t = 9/5$:

$$\mathbf{m} = \begin{pmatrix} 3 - 3(9/5) \\ 9/5 \end{pmatrix} = \begin{pmatrix} 15/5 - 27/5 \\ 9/5 \end{pmatrix} = \begin{pmatrix} -12/5 \\ 9/5 \end{pmatrix} \quad (8)$$

The area of a triangle with vertices at the origin **O**, **A**, and **M** is given by half the absolute value of the determinant of the matrix formed by the position vectors **a** and **m**.

$$\text{Area} = \frac{1}{2} \left| \det \begin{pmatrix} \mathbf{a} & \mathbf{m} \end{pmatrix} \right| = \frac{1}{2} \left| \det \begin{pmatrix} 3 & -12/5 \\ 0 & 9/5 \end{pmatrix} \right|$$

$$\text{Area} = \frac{1}{2} \left| \left(3 \cdot \frac{9}{5} \right) - \left(0 \cdot \frac{-12}{5} \right) \right| = \frac{1}{2} \left| \frac{27}{5} \right| = \frac{27}{10}$$

The area of the triangle OAM is $\frac{27}{10}$. This corresponds to option (d).

